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LUBRICATION ASSEMBLY AND METHOD FOR USE THEREOF

Abstract

A system and method for assisting a surgeon performing an ophthalmic procedure on an eye of a patient, the system including a lubrication assembly. The lubrication assembly may include a mounting bracket, a dispenser, a lubricant source, a fluid line, and a first actuator. The dispenser is coupled to the mounting bracket which is configured to be coupled to a surgical microscope. The lubricant source is fluidly coupled to the dispenser by the fluid line for movement of a lubricant between the lubricant source and the dispenser. The first actuator dispenses the lubricant from the dispenser onto the eye of the patient.

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Background/Summary

INTRODUCTION

[0001] Ophthalmic or ocular surgeries require frequent lubrication of a patient's eye while being performed. The eye of the patient must be constantly lubricated during all types of ophthalmic procedures so as to avoid damage to ocular tissue. Traditionally, while the surgeon is performing the ophthalmic procedure, an assistant or other medical professional is within the vicinity with a dropper or syringe filled with lubricant to manually disperse the lubricant. When the surgeon calls out for more lubricant, the assistant then manually applies it to the patient's eye using the dropper.

SUMMARY

[0002] The present disclosure relates generally to a lubrication assembly for surgical microscopes and a method for use thereof.

[0003] In certain embodiments, a lubrication assembly for lubricating an eye of a patient during an ophthalmic procedure is provided. The lubrication assembly includes a mounting bracket configured to be coupled to a surgical microscope, a dispenser coupled to the mounting bracket, a lubricant source fluidly coupled to the dispenser, a fluid line for movement of a lubricant between the lubricant source and the dispenser, and a first actuator for dispensing, from the dispenser, the lubricant onto the eye of the patient.

[0004] In certain embodiments, a method for lubricating an eye of a patient during an ophthalmic procedure is provided. The method includes moving a lubrication assembly into a selected position relative to the eye of the patient, dispensing a lubricant from the lubrication assembly onto the eye of the patient, and monitoring a position of the lubrication assembly relative to the eye of the patient for the duration of the ophthalmic procedure.

[0005] The following description and the related drawings set forth in detail certain illustrative features of one or more embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The appended figures depict certain aspects of one or more disclosed embodiments and are therefore not to be considered limiting of the scope of this disclosure.

[0007] FIG. 1 illustrates a surgical setting, according to certain embodiments.

[0008] FIG. 2 is a schematic diagram of a lubrication assembly, according to certain embodiments.

[0009] FIG. 3A is a magnified cross-sectional side view depicting the lubrication assembly of FIG. 2 coupled to a surgical microscope, according to certain embodiments.

[0010] FIG. 3B is a bottom-up view of the surgical microscope and the lubrication assembly of FIG. 3A, according to certain embodiments.

[0011] FIG. 4 is a schematic diagram illustrating components of the surgical system, according to certain embodiments.

[0012] FIG. 5 is a flow diagram illustrating a technique for lubricating an eye of a patient during an ophthalmic procedure, according to certain embodiments.

[0013] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the drawings. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0014] It will be readily understood that the components of the embodiments as generally described herein and illustrated in the appended Figures can be arranged and designed in a wide variety of

different configurations. Thus, the following more detailed description of various embodiments, as represented in the Figures, is not intended to limit the scope of the present disclosure but is merely representative of various embodiments. While the various aspects of the embodiments are presented in the Figures, the Figures are not necessarily drawn to scale unless specifically indicated.

[0015] As used herein, the term “surgical setting” may refer to any environment where surgical procedures are performed. For example, the term “surgical setting” may refer to an operating theater or an operating room, with one or more surgeons and operating staff involved in the surgical setting.

[0016] As used herein, the term “surgical system” may refer to any surgical system, console, or device for performing a surgical procedure. For example, the term “surgical system” may refer to a surgical tool or system, such as a phacoemulsification console, a laser system, an imaging system, an intraocular lens (IOL) alignment system, a biometer, an optical coherence tomography (OCT) machine, or a vitrectomy console.

[0017] As used herein, the term “distal” may refer to a system, device, component, end, portion, or segment that is disposed closer to a patient and/or further from a console during an ophthalmic procedure; and the term “proximal” refers to the system, device, component, end, portion, or segment that is disposed further from the patient and/or closer to the console during the ophthalmic procedure.

[0018] Although generally described with reference to an ophthalmic surgical setting, the devices and systems described herein may be implemented in other settings and contexts, such as other surgical settings, without departing from the scope of the present application.

[0019] Manual application of lubricant to a patient's eye as described above can carry with it several disadvantages. For example, the application of lubricant by the assistant or other medical professional can temporarily block or obstruct the surgeon's view of the patient's eye, leading to inefficient performance of the ophthalmic procedure. Additionally, the assistant applying the lubricant may apply either too much, too little, or may apply lubricant to the wrong portion of the patient's eye, thereby leading to inefficient use of the lubricant, which can be expensive and therefore increase the overall cost of the ophthalmic procedure. Furthermore, because the assistant or surgeon must constantly judge the lubrication state of the patient's eye, this can lead to too much elapsed time between lubrication applications, which can make the ophthalmic procedure more uncomfortable for the patient than is necessary. Alternatively, if too much lubrication is applied, this can cause excessive runoff and obscure the surgeon's view at critical points during a surgical procedure.

[0020] Accordingly, the systems described herein may overcome many of the limitations associated with current lubrication techniques.

[0021] Certain embodiments described herein provide improved lubrication assemblies and techniques for use thereof during the performance of ophthalmic procedures. More particularly, certain embodiments provide lubrication assemblies that automatically dispense a lubricant or other surgical fluid onto an eye of a patient, which eliminates the need for an assistant to manually dispense lubricant onto the eye and reduces complications associated with insufficient lubricant, or other similar events affecting the eye of the patient during ophthalmic procedures.

[0022] FIG. 1 illustrates an example of a surgical setting **100** in which a surgeon **132** and a patient **130** are present, according to certain embodiments of the present disclosure. The patient **130** is partially covered by a surgical drape **194** and positioned on a patient support table **192**, which is illustrated as fixed to the floor of the surgical setting **100** in which the procedure is performed. Although one surgeon **132** is illustrated in FIG. 1, multiple surgeons and/or operating staff may also be present within the surgical setting **100**. For example, in certain embodiments, a surgical assistant and/or circulating nurse (i.e., circulator) may also be present within the surgical setting **100**.

[0023] As shown, the surgeon **132** is operating a surgical microscope system **110** as part of a procedure on the patient **130**. Examples of the procedure may include ophthalmic procedures such as, cataract surgeries, vitrectomy surgeries, and/or glaucoma surgeries, but may also include other ophthalmic procedures that involve lubrication or hydration of the eye. The surgical microscope system **110** includes a surgical microscope **102** (e.g., ophthalmic surgical microscope) with a lubrication assembly **104** coupled thereto, an arm **106**, a base **108**, and video display monitors **112a**, **112b**, and **112c**. The surgical microscope **102** is moveably attached to the arm **106**, which is moveably attached to the base **108**. As such, the surgeon **132** can manually maneuver the surgical microscope **102** and the lubrication assembly **104** relative to the patient **130** via manipulation of the surgical microscope **102** and/or the arm **106**.

[0024] The surgical microscope **102** is operable to hold, move, and position a lubrication assembly **104** in the surgical setting **100** and within a close proximity to an eye **180** of the patient **130** (best seen in FIG. 3A). Although it is currently contemplated that the patient will most frequently be a human patient, in veterinary applications, the “patient” can be a non-human animal, and the term as it appears in this disclosure should be read to encompass either possibility. Additionally, while the surgical microscope **102** is attached to the arm **106** that includes a series of components as shown and described, in certain other embodiments, a robotic arm or other apparatus can be used. For example, rather than a maneuverable arm, a series or plurality of cascaded linear rails may be utilized to secure, move, and provide lubrication assembly **104** to the surgical setting **100** and within close proximity to the eye **180** of the patient **130**.

[0025] The arm **106** may be supported by any suitable device or system within an operating environment. In the example of FIG. 1, the arm **106** is supported by the base **108**. The base **108** can itself be fixed or movable with respect to a structure for supporting the patient **130**. For example, the base **108** may be supported by lockable wheels **114** that allow the base **108** to be rolled into position with respect to the patient support table **192**. In certain embodiments, however, the arm **106** may be supported by the surgical console **120**, the patient support table **192**, or another supporting structure within the operating environment. For example, the arm **106** may be movably attached to a supporting mount on a ceiling of the surgical setting **100**. Generally, the arm **106** may be coupled to the surgical microscope **102** so that movement of the arm **106** may manipulate both the surgical microscope **102** and the lubrication assembly **104** at the same time.

[0026] The configuration of the arm **106** and the base **108** in FIG. 1 is only exemplary, and the exact configuration of the arm **106** and the base **108** can vary considerably in any given embodiment. For example, in certain embodiments, the arm **106** includes elements that provide the arm **106** movement with at least six degrees of freedom. The video display monitors **112** may deliver information and images to medical personnel (e.g., the surgeon **132** and/or the assistant(s)) during the ophthalmic procedure. FIG. 1 illustrates a system with a first display monitor **112a** disposed on the arm **106**, a second display monitor **112b** disposed between the arm **106** and the surgical microscope **102**, and a third display monitor **112c** disposed on the base **108** for use, for example, by the surgeon **132** and/or the assistant(s). The display monitors **112** can be communicatively coupled with the surgical microscope **102**, the surgical console **120**, and/or a visualization system within the operating environment. In some embodiments, one or more of the display monitors **112** can receive information (e.g., surgical parameters) and/or images from the surgical console **120**, and display the information and images on the one or more display monitors **112**. The surgical console **120** can also send signals to the display monitor **112** for performing operations (e.g., starting and stopping video recording).

[0027] The surgical microscope system **110** may be in direct or indirect communication with one or more surgical systems, consoles, and/or devices (e.g., integrated in an inter-networked surgical suite) within the surgical setting **100**, such as the lubrication assembly **104**, the video display monitors **112**, and/or the surgical console **120**. Examples of suitable surgical systems that may be included in the surgical suite include surgical consoles for performing vitreoretinal procedures,

cataract surgeries, corneal transplants, glaucoma surgeries, LASIK surgeries, refractive lens exchanges, trabeculectomies, and refractive surgeries, among other consoles, imaging devices, laser devices, diagnostic devices, and accessories identifiable by those of ordinary skill.

[0028] In certain embodiments, the surgical microscope system **110** is a stand-alone device or module that is in wireless or wired communication with the one or more surgical systems physically positioned within the surgical setting **100**, such as the surgical console **120**. In certain other embodiments, however, the surgical microscope system **110** is integrated within one or more of the surgical systems physically positioned within the surgical setting **100**. For example, the surgical microscope system **110** may be integrated with the lubrication assembly **104**, the video display monitors **112**, and/or the surgical console **120**.

[0029] FIG. **2** is a schematic diagram of the lubrication assembly **104** of FIG. **1**, according to certain embodiments. The lubrication assembly **104** includes a mounting bracket **140**, a dispenser **142**, a lubricant source **146**, a fluid line **144**, and an optional lubricant actuator **148** (also referred to as a “first actuator” herein).

[0030] The mounting bracket **140** is configured to couple to the surgical microscope **102** shown in FIG. **1** and attach the dispenser **142** and fluid line **144** thereto. In certain embodiments, the mounting bracket **140** is a dovetail bracket or other suitable rail-type bracket configured to mount to a head **105** of the surgical microscope **102** (best seen in FIG. **3A**). The mounting bracket **140** may also be configured to mount to the surgical microscope **102** via one or more screws, an adhesive, or other similar mounting feature. Generally, the mounting bracket **140** is configured to hold the dispenser **142** and the fluid line **144** coupled to the dispenser when the mounting bracket **140** is attached to the surgical microscope **102** such that the dispenser **142** and the fluid line **144** do not obstruct an ophthalmic procedure being performed by the surgeon **132**. Generally, a position of the mounting bracket **140** is adjustable, or the position of the dispenser **142** within the mounting bracket **140** is adjustable, such that a position of the dispenser **142** may be manually or automatically adjusted relative to the surgical microscope before or during an ophthalmic procedure.

[0031] The dispenser **142** is configured to dispense a lubricant (e.g., a balanced salt solution (BSS)) or other surgical fluid (e.g., a numbing fluid or a dilating fluid) onto the eye **180** of the patient **130**. In certain embodiments, the dispenser **142** is a syringe needle. In other embodiments, the dispenser **142** may be an eye dropper, pipette, tubing, nozzle, spout, or other type of fluid conduit configured to dispense fluid in a precise and controlled manner. The dispenser **142** is coupled to the mounting bracket **140** at a connection end **164** of the dispenser **142** and dispenses the lubricant from a dispensing end **166** of the dispenser **142**. In certain embodiments, the dispensing end **166** is angled relative to an axis **156** of the dispenser **142** by an angle **152**. In certain embodiments, the axis **156** is a major longitudinal axis of the dispenser **142** that extends substantially horizontally relative to a floor of the surgical setting **100** when the dispenser **142** is attached to the surgical microscope **102** via the mounting bracket **140**. The angle **152** may be an acute angle (e.g., between 0-90 degrees), a right (e.g., 90 degrees) angle, or an obtuse (e.g., greater than a 90 degrees) angle. The angled orientation of the dispensing end **166** of the dispenser **142** may help facilitate dispensing or dripping of the lubricant onto the eye **180** by preventing the lubricant from collecting at the dispensing end **166**, or within the dispenser **142** generally. In other embodiments, however, the dispensing end **166** is disposed along the same longitudinal axis as the connection end **164**. Or stated another way, in such embodiments, the angle **152** between the dispensing end **166** and the axis **156** is 0 degrees.

[0032] The fluid line **144** is configured to facilitate movement of (or flow of) the lubricant between the lubricant source **146** and the dispenser **142**. In other words, the fluid line **144** acts as a conduit for the lubricant. In some embodiments, the fluid line **144** is a cable or other similar tubing provided for supplying the lubricant and/or other fluids, such as water, directly to the dispenser **142** of the lubrication assembly **104**. As shown in FIG. **2**, a distal end **163** of the fluid line **144** is

fluidically coupled to the dispenser **142**, while an opposing proximal end **165** of the fluid line **144** is fluidically coupled to the lubricant source **146** or other lubricant reservoir. The fluid line **144** may be at least partially disposed within the mounting bracket **140**, and in certain embodiments, may be connected to the connection end **164** of the dispenser **142** within the mounting bracket **140**. In certain embodiments, the fluid line **144** is completely disposed exterior to the surgical microscope system **110**, or at least partially disposed exterior to the surgical microscope system **110**. For example, a portion of the fluid line **144** near the mounting bracket **140** may be disposed exterior to the surgical microscope system **110**, and a portion of the fluid line **144** connected to the lubricant source **146** may be disposed within a housing of the surgical microscope **102**, the arm **106**, and/or the base **108**. Generally, the lubricant source **146** may be disposed external to the microscope system **110**, and in some examples, attached with one or more components of the microscope system **110**, or the lubricant source **146** may be integrated with one or more components of the microscope system **110**.

[0033] In certain embodiments, the lubrication assembly **104** includes a controlled release valve **151** disposed at the connection end **164** of the dispenser **142**, a distal end of the lubricant source **146**, and/or at any point of the fluid line **144** or connection elements disposed therebetween to facilitate controllable flow of lubricant from the lubricant source **146** to the dispenser **142**, and/or controllable dispensing of the lubricant from the dispenser **142**. The controlled release valve **151** is exemplarily shown near the connection end **164** of the dispenser **142** in FIG. 2. The controlled release valve **151** is particularly useful in embodiments of the lubrication assembly **104** without the lubricant actuator **148**, where dispensing is facilitated by gravity feed, to control a dispense rate of the gravity feed lubrication assembly **104**. Even in embodiments with the lubricant actuator **148**, the controlled release valve **151** may be beneficially used to prevent any unwanted dispensing, or leaking, of lubricant from the dispenser **142** when the controlled release valve **151** is disposed at or near the dispenser **142**.

[0034] In certain embodiments, the controlled release valve **151** is manually controlled. In certain embodiments, the controlled release valve **151** is electronically controlled. In certain embodiments, the controlled release valve **151** may be adjusted (e.g., between an opened and closed position) to allow lubricant to flow from the lubricant source **146** to the dispenser **142**, and/or from the dispenser **142** onto the patient's eye **180**, at a target flow and/or dispense rate, respectively. The target flow and/or target dispense rate referred to above may be defined by a user. For example, the controlled release valve **151** may be adjusted to allow a target volume (e.g., 1 to 10 milliliters) of the lubricant to be dispensed, in droplets, from the dispenser **142** at a target dispense rate of, for example, once every 1-60 seconds (e.g., once every 1 second, 3 seconds, 5 seconds, 10 seconds, etc.) or once every more than 60 seconds (e.g., once every 90 seconds, 120 seconds, 150 seconds, etc.), or any range associated therewith (e.g., once between very 1 second and 10 seconds, etc.).

[0035] In certain embodiments, a level of lubricant within the lubricant source **146** is maintained by manually refilling or replenishing a refillable cartridge **153** or other containment device inserted into the lubricant source **146**, or by a direct injection of additional lubricant by the surgeon **132** or the assistant through a port **155** defined on the lubricant source **146** via a separate syringe. In some embodiments, the port **155** may include a self-sealing port, such as a silicone gel or other material. In certain embodiments, the lubricant source **146** is pressure regulated via the lubricant actuator **148** so as to provide a consistent and even flow of the lubricant through the dispenser **142**. That is, the lubricant actuator **148** maintains an internal pressure in the lubricant source **146** during dispensing of the lubricant.

[0036] The lubricant actuator **148** may be operably coupled to the lubricant source **146** and configured to expel lubricant from the lubricant source **146** to the dispenser **142** via the fluid line **144**. That is, the lubricant actuator **148** is configured to act upon the lubricant source **146** so as to force the lubricant from the lubricant source **146** to the dispenser **142**, and out of the dispensing end **166** onto the eye **180**. In certain embodiments, the lubricant actuator **148** includes a pump, such

as a positive displacement pump, for expelling the lubricant from the lubricant source **146** to the dispenser **142**. For example, the pump may include a diaphragm pump, a peristaltic pump, a gear pump, a piston pump, or the like. In certain embodiments, the lubricant actuator **148** includes another type of mechanical compression or squeezing device, or other similar actuator. Although the lubricant actuator **148** is shown adjacent to the lubricant source **146** in FIG. 2, the lubricant actuator **148** may be disposed within the lubricant source **146**.

[0037] In certain embodiments, the lubricant actuator **148** is configured to drive lubricant from the lubricant source **146** to the dispenser **142** such that the lubricant is dispensed from the dispenser **142** at a target dispense rate. For example, a target volume (e.g., 1 to 10 milliliters) of the lubricant may be dispensed, in droplets, from the dispenser **142** at a target dispense rate of, for example, once every 1-60 seconds (e.g., once every 1 second, 3 seconds, 5 seconds, 10 seconds, etc.) or once every more than 60 seconds (e.g., once every 90 seconds, 120 seconds, 150 seconds, etc.), or any range associated therewith (e.g., once between every 1 second and 10 seconds, etc.). In certain embodiments, the lubricant actuator **148** is configured to drive lubricant from the lubricant source **146** to the dispenser **142** at a predetermined, or preset, dispense rate that may be determined automatically or via user input. In certain embodiments, the dispense rate is based on a type of ophthalmic procedure being performed by the surgeon **132**. That is, a faster dispense rate may be used for procedures that require more lubrication, and a slower dispense rate may be used for procedures that require less lubrication. In certain embodiments, the dispense rate is based on a size of the eye **180**. In further embodiments, the lubricant actuator **148** may be programmed or configured to drive lubricant from the lubricant source **146** to the dispenser **142** such that the lubricant is continuously dispensed from the dispenser **142** in a continuous stream.

[0038] In certain embodiments, the dispense rate when using the lubricant actuator **148** may be set (e.g., preset) by a user (e.g., the surgeon **132** or the assistant) or a corresponding controller, and can be changed during the ophthalmic procedure to adjust the amount of lubrication dispensed onto the eye **180**. Dispensing of the lubricant may be manually controlled via various types of user input devices, for example via a manual knob or a valve (e.g., the controlled release valve **151**) disposed on the surgical microscope **102** or the surgical console **120** or the lubrication assembly **104**, a graphical user interface of the surgical microscope system **110** (e.g., one or more of the display monitor(s) **112**) or the surgical console **120**, a foot pedal in wired or wireless communication with the lubrication assembly **104** and/or the surgical console **120**, or other analog or digital input devices such as a button, switch, keyboard, or the like. In certain embodiments, dispensing of the lubricant may be controlled via audible commands issued by the surgeon **132** or the assistant using known voice command means (e.g., devices and corresponding software) in communication with the lubrication assembly **104**. In certain embodiments, the dispense rate is automatically controlled in response to a determined dryness and/or wetness of the eye **180** as described with reference to FIG. 4.

[0039] In certain embodiments, dispensing of the lubricant from the lubrication assembly **104** is dictated by a predetermined or time-based schedule. For example, the surgical console **120** or the surgical microscope system **110** may include a clock timer and/or related software that allows the surgeon **132** or the assistant to establish a time interval between each droplet or instance of a predetermined amount of lubricant being dispensed from the lubricant assembly **104**. In such embodiments, the lubricant assembly **104** can automatically dispense lubricant according to the established time interval for a predetermined duration of the ophthalmic procedure. If the surgeon **132** or assistant determines that either too much or too little lubricant is being applied, they may adjust the time interval, or the predetermined amount of lubricant being dispensed during each time interval, accordingly until an appropriate amount of lubricant is being applied to the patient's eye **180**.

[0040] FIG. 3A is a magnified cross-sectional side view depicting the lubrication assembly **104** of FIG. 2 coupled to the surgical microscope **102** of FIG. 1, during use, according to certain

embodiments. As shown in FIG. 3A, the surgical microscope **102** and the lubrication assembly **104** are positioned in close proximity to the eye **180** of the patient **130**, for example, during an ophthalmic procedure. FIG. 3B is a view from the eye **180** of the patient **130** up towards the surgical microscope **102** and the lubrication assembly **104** of FIG. 3A, according to certain embodiments. FIGS. 3A-3B are described together herein for clarity purposes.

[0041] In certain embodiments, the surgical microscope **102** includes a head **105**, an optic system **115** with optics **117**, a window (or other optical component) **103**, a camera **160** (illustrated in FIG. 3B), and an illuminator **162** (illustrated in FIG. 3B). The optic system **115** with optics **117** is disposed within the surgical microscope **102**, and the window **103** is centrally disposed at a bottom of the head **105**. The camera **160** and the illuminator **162** are disposed radially outward of the window **103** in positions along the head **105** that are unobstructed by the lubrication assembly **104**. Although the window **103**, the camera **160**, and the illuminator **162** are shown in a particular arrangement along the head **105** in the present examples, the window **103**, the camera **160**, and the illuminator **162** may be positioned in other arrangements in other examples. Further, in certain embodiments, one or both of the camera **160** and the illuminator **162** may be separate devices or components from the surgical microscope **102**, and may be attached to the surgical microscope **102** or disposed nearby the head **105** of the surgical microscope **102** in the surgical setting.

[0042] The optic system **115** is configured to enable the surgeon **132** using the surgical microscope **102** to observe the eye **180** of the patient **130** at increased magnification and/or under other conditions. For example, the optic system **115** may provide the surgeon **132** with a magnified view of the eye **180** that can be adjusted via manipulation of the optics **117** of optic system **115**. In certain embodiments, the optic system **115** is shielded at the bottom of surgical microscope **102** by the window **103**, which is configured to prevent debris or other external materials from penetrating and/or affecting the optic system **115** and/or the optics **117**. In certain embodiments, the window **103** is transparent and/or coated with an optical coating to enhance transmission.

[0043] A field of view (FOV) **111** of the surgical microscope **102**, as indicated by the dashed region, may be seen through an eyepiece of the surgical microscope **102** and/or on one or more of the display monitor(s) **112** (FIG. 1). The FOV **111** defines a maximum area or visual field of the optics **117** of the optic system **115**. In certain embodiments, a focal plane **107** of the surgical microscope **102**, as determined by a focal length of the optics system **115**, can be manipulated by the surgeon **132** to move a point of focus along a longitudinal axis **109** of the surgical microscope **102**, which may be an optical axis of the optics **117** and/or optic system **115**. The focal plane **107** is perpendicular to the longitudinal axis **109** and defines a distance from the window **103** at which an object (e.g., the eye **180**) comes into focus.

[0044] The illuminator **162** generates and/or emits visible light or other desired wavelengths of light from the head **105** for illuminating the eye **180** or other locations of interest during an ophthalmic procedure. Illumination light reflected from the eye **180** returns to the camera **160**, which captures images of the eye **180** and/or dispenser **142** of the lubrication assembly **104**. The camera **160** then transmits the images to other components of the surgical microscope system **110** and/or surgical console **120** for processing and, in some embodiments, for display, e.g., as still images or continuous video, on one or more of the display monitor(s) **112**, where they can be seen and used by the surgeon **132**, an assistant, and/or others involved in the procedure.

[0045] In certain embodiments, the images captured by the camera **160** may, for example, be used to continuously track or monitor a position of the eye **180** of the patient **130**, the position of the lubrication assembly **104** (i.e., the dispensing end **166**) relative to the eye **180** of the patient **130**, a lubrication level of the eye **180**, and/or to assist in the targeted lubrication of the patient's eye **180**, as is further discussed below. In certain embodiments, continuous tracking or monitoring of the position of the eye **180** is carried out via tracking or monitoring of a position of a pupil or vasculature of the eye **180**. Although the camera **160** is shown as being disposed through the bottom of surgical microscope **102** in FIG. 3B and configured to directly receive images of the

illuminated eye **180**, in certain embodiments, the camera is integrated inside surgical microscope **102** or is in optical communication with one or more optics **117** of the optic system **115**, and is configured to receive images of the eye **180** via the optics **117**.

[0046] The lubrication assembly **104** includes the mounting bracket **140**, the dispenser **142**, the lubricant source **146**, the fluid line **144**, and in certain embodiments, the lubricant actuator **148**, as described with reference to FIG. 2. When the lubrication assembly **104** is mounted to the surgical microscope **102** via the mounting bracket **140**, the dispenser **142** and/or the mounting bracket **140** may be adjusted such that the dispensing end **166** extends into the FOV **111**, and in certain examples, is positioned along the longitudinal axis **109** between the surgical microscope **102** and the eye **180** (e.g., at or near a center **113** of the window **103**). Although the dispenser **142** is disposed in the FOV **111**, the dispenser **142** is positioned outside of the working range of focal planes **107** of the surgical microscope **102**, such that the dispenser **142** does not obstruct the surgeon **132**'s view of the eye **180** during the ophthalmic procedure. In other words, when the surgeon **132** is examining the eye **180** through the surgical microscope **102**, the dispenser **142** is not visible to the surgeon **132**.

[0047] When the dispensing end **166** of the dispenser **142** is positioned along longitudinal axis **109**, lubricant **190** can be dispensed directly onto an area of the eye **180** on or through which the surgeon **132** is examining the performance of an ophthalmic procedure. Alternatively, when the dispensing end **166** of the dispenser **142** is positioned adjacent to or away from the longitudinal axis **109**, lubricant **190** can be dispensed onto the eye **180** at a position adjacent to or away from the area of the eye **180** on or through which the surgeon **132** is examining. When continuous lubrication of eye **180** is desired at the area of the eye **180** on or through which the surgeon **132** is viewing, the lubrication assembly **104** allows the surgeon **132** to continue to examine and/or perform the ophthalmic procedure without visual interruptions for lubrication. For example, the surgeon **132** does not have to stop and wait for the assistant to manually lubricant the eye **180**, during which the assistant may also obstruct the surgeon **132**'s view of the eye **180**. The lubrication assembly **104** therefore helps reduce interruptions during ophthalmic procedures and reduces complications associated with insufficient lubrication of the eye **180**.

[0048] In certain embodiments, the lubrication assembly **104** may further include a movement actuator **154** (which may also be referred to as a “second actuator”) for automatically moving the dispensing end **166** of the dispenser **142** into a selected position relative to the eye **180**, as opposed to manual adjustment by a user. The selected position referred to above may be a position that is selected by the user. The movement actuator **154** may include a motor, a drive assembly, or other similar actuator for translating the dispenser **142**. In certain embodiments, the movement actuator **154** is configured to linearly move the dispensing end **166** of the dispenser **142** into the selected position by moving the mounting bracket **140** along a y-axis **157** and/or an x-axis **159** (shown in FIG. 3B). As such, the dispensing end **166** may be moved away or toward longitudinal axis **109**. In certain embodiments, the movement actuator **154** is coupled to the head **105** of the surgical microscope **102** and the mounting bracket **140** of the lubrication assembly **104**. In certain embodiments, the movement actuator **154** is movably coupled to the head **105** and is fixed to the mounting bracket **140**, or vice-versa.

[0049] Movement of the dispenser **142** via the movement actuator **154** may be controlled via user input by the surgeon **132** or other operating staff (e.g., on the surgical console **120**, surgical microscope system **110**, or other input device), or automatically by a controller of the lubrication assembly **104**, surgical microscope system **110**, or surgical console **120** in communication therewith. For example, the surgeon **132** can manually control the movement actuator **154** via a knob on the surgical microscope **102** or the lubrication assembly **104**, a graphical user interface of the surgical microscope system **110** (e.g., one or more of the display monitor(s) **112**), a foot pedal, or any other analog or digital input device such as a button, switch, keyboard, or the like, or by audible commands issued by the surgeon **132** or the assistant using known voice command means.

Alternatively, or in combination therewith, actuation of the movement actuator **154** may be automatically controlled by comparison of the position of the surgical microscope **102**, the lubrication assembly **104**, and/or the dispensing end **166** with the position of the eye **180**, as determined by the surgical microscope system **110** or surgical console **120** in communication therewith.

[0050] In certain embodiments, the lubrication assembly **104** may be fixedly attached to the head **105** of the surgical microscope **102**. In many cases, though, it may be preferable that the lubrication assembly **104** be conveniently removable and replaceable from the head **105** of the surgical microscope **102**. A lubrication assembly **104** that is removable with respect to the surgical microscope allows for the replacement of a malfunctioning lubrication assembly **104**, for the removal and sterilization of a lubrication assembly **104** between procedures, for the removal and disposal of a single-use (e.g., disposable) lubrication assembly **104** after a procedure, and/or for other advantages as appropriate depending on the particular application.

[0051] FIG. **4** is a schematic system diagram illustrating various example components of the lubrication assembly **104**, and the associated surgical microscope system **110** and surgical console **120** of FIG. **1**. In the example of FIG. **1**, the base **108** provides physical support for the arm **106**, which in turn carries the surgical microscope **102** and the lubrication assembly **104**, which may be in wired or wireless communication with the surgical console **120**. As shown in FIG. **4**, the lubrication assembly **104** includes an optional lubrication assembly controller **402**, which may be integrated with or in wired or wireless communication with the other components of lubrication assembly **104**; the surgical microscope includes an optional surgical microscope controller **404**, which may be integrated with or in wired or wireless communication with the other components of surgical microscope **102**; and the surgical console **120** includes a surgical console controller **406**, which may be integrated with or in wired or wireless communication with the other components of the surgical console **120**. The surgical microscope **102** and/or the surgical console **120** are also in wireless or wired communication with the display **112**. Note that although certain functions below are described with reference to the surgical console controller **406**, such functions may be performed by one or both of the lubrication assembly controller **402** and the surgical microscope controller **404**, separately from or in combination with the surgical console controller **406**.

[0052] Generally, each of the controllers **402**, **404**, and/or **406** include a processor and a memory. The memory may include any device operable to receive, store, or recall data, including, but not limited to, electronic, magnetic, or optical memory, whether volatile or non-volatile. The memory may include code stored thereon. The code may include instructions that may be executable by the processor. The code may be created, for example, using any programming language, including but not limited to, C++ or any other programming language (including assembly languages, hardware description languages, and database programming languages).

[0053] The processor may be, or include, a microprocessor, a microcontroller, an embedded microcontroller, a programmable digital signal processor, or any other programmable device operable to receive information from the memory or other devices in communication with the processor and/or controllers **402**, **404**, **406**, and perform one or more operations on the received information. For example, the processor may send instructions to components of the surgical microscope system **110**, or other devices or systems in communication with controllers **402**, **404**, and/or **406**, for controlling such devices and systems. The processor may also be operable to output results based on the operations performed thereby. In certain examples, display **112** shows data and other output results provided by the processor of a controller. In some instances, the processor may also be or include an application specific integrated circuit, a programmable gate array, programmable array logic, or any other device or combinations of devices operable to process electric signals.

[0054] As shown in FIG. **4**, the surgical microscope **102** further includes the illuminator **162**, which generates and delivers illumination energy to the eye **180** of the patient **130** to illuminate the eye

180, and the camera **160**, which is configured to capture images of the illuminated eye **180**. The illumination energy from the illuminator **162** may often be in the form of visible light, but may also include near-infrared or ultraviolet energy, or other illumination energies (including those from laser sources) appropriate to the visualization desired for a given application. The illumination energy projected from the illuminator **162** is reflected from the patient's eye **180** back to the camera **160**. This reflected energy may be captured (e.g., converted to digital signals) by one or more sensors of the camera **160**. The camera **160** may include, for example, an active-pixel sensor based on a complementary metal-oxide semiconductor (CMOS) sensor.

[0055] The digital signals corresponding to images (e.g., image data) captured by the camera **160** may be transmitted to the surgical console controller **406**, or other suitable controller, where image analysis software contained therein continuously analyzes the digital signals. In certain embodiments, the image analysis software analyzes the digital signals to determine and/or monitor a lubrication level (e.g., a relative amount of moisture or lubrication) on the surface of the eye **180** of the patient. For example, the image analysis software may determine and/or monitor the general lubrication level of the entire surface of the eye **180**, or may determine and/or monitor the lubrication level of one or more particular regions of the eye **180**. To determine lubrication levels, the image analysis software can detect and/or monitor amounts of glare within the transmitted image(s), where the amounts of glare correspond to a lubrication level. The glare can be detected and/or monitored for the entire surface of the eye **180**, or one or more particular regions of the eye **180**. Particular regions of the eye **180** can be identified and monitored by determining a relative position of the eye **180** via tracking of a pupil or vasculature of the eye **180**.

[0056] In certain embodiments, in response to a determination of a lubrication level, the surgical console controller **406** can send a signal to the lubrication assembly **104** to automatically adjust a dispense rate of lubricant. For example, in certain embodiments, when a detected level (for a particular region or entire surface of the eye **180**) of glare drops below a predetermined threshold value, thereby indicating a low level of moisture or lubricant on the surface of the patient's eye **180**, the surgical console controller **406** sends a signal to the lubricant actuator **148**, or controlled release valve **151**, to increase a dispense rate of lubricant. The signal may cause the lubricant actuator **148** to pump lubricant from the lubricant source **146** faster, or may cause the controlled release valve **151** to further open. Conversely, when a large amount of glare has been detected, thereby indicating that the patient's eye **180** has a sufficient amount of lubrication, the surgical console controller **406** sends no signal to the lubricant actuator **148** or controlled release valve **151**, or sends a signal to the lubricant actuator **148** or controlled release valve **151** to reduce a dispense rate of lubricant. The signal may cause the lubricant actuator **148** to stop pumping or pump lubricant from the lubricant source **146** slower, or may cause the controlled release valve **151** to further close.

[0057] In certain embodiments, the surgical console controller **406** may continuously monitor the amount of glare within the received image data so that lubricant may be added in real-time to the patient's eye **180** during the ophthalmic procedure; however in other embodiments, the surgical console controller **406** periodically detects the amount of glare within the received image data according to a predetermined schedule or time interval so that lubricant is only periodically added during selected moments of the ophthalmic procedure.

[0058] In certain embodiments, in response to a determination of a lubrication level, the surgical console controller **406** can also send a signal to the lubrication assembly **104** to automatically adjust a position of the dispensing end **166** of the lubrication assembly **104**. For example, in certain embodiments, when a detected level of glare for a particular region of the eye **180** drops below a predetermined threshold value, the surgical console controller **406** sends a signal to the movement actuator **154** to move the lubrication assembly **104** such that the dispenser **142** of the lubrication assembly **104** is disposed over the particular region.

[0059] In certain embodiments, instead of automatically adjusting a dispense rate of lubricant from the lubrication assembly **104** or a position of the dispensing end **166** of the lubrication assembly

104 in response to a determination of a lubrication level, the surgical console controller **406** can provide a notification to the surgeon or other operating staff to adjust the dispense rate or reposition the dispensing end **166**. For example, in certain embodiments, the surgical console controller **406** may generate and transmit a visual signal to the display **112** to display a graphic or text message for inspection by the surgeon or other operating staff. In certain embodiments, the surgical console controller **406** may generate and transmit an audio signal to a speaker of the surgical console **120**, surgical microscope **102**, and/or lubrication assembly **104** to emit an audible tone or message for the surgeon or other operating staff.

[0060] In certain embodiments, in addition to or separately from monitoring lubrication levels, the surgical console controller **406**, or other suitable controller, can monitor or track the position of the dispensing end **166** of the lubrication assembly **104** relative to the patient's eye **180** or portions thereof. For example, the surgical console controller **406** may comprise image analysis software that analyzes digital signals received from the camera **160** to determine and/or monitor a position of the lubrication assembly **104** and thus, a position of the dispensing end **166** of the dispenser **142**, relative to the position of the eye **180**. Determinations of the position of the dispensing end **166** may be made by comparing a position of, e.g., the dispensing end **166** over the eye **180** and the positions of a pupil or vasculature of the eye **180**. In certain embodiments, a relative position of the dispensing end **166** is determined via one or more positional encoders in the lubrication assembly **104**.

[0061] Monitoring or tracking the position of the dispensing end **166** of the lubrication assembly **104** can be used to maintain a selected position of the lubrication assembly **104** relative to the eye **180**, or for modifying an amount of lubricant being applied to a particular region of the eye **180**, as described below. In certain embodiments, in response to a determination that the position of the dispensing end **166** of the lubrication assembly **104** over the eye **180** has changed relative to a selected position, the surgical console controller **406** can send a signal to the lubrication assembly **104** to automatically adjust a position of the dispensing end **166**. For example, in certain embodiments, when a detected position of the dispensing end **166** is different from a preset or pre-programmed position, the surgical console controller **406** sends a signal to the movement actuator **154** to adjust the detected position to the preset position.

[0062] In certain embodiments, the surgical console controller **406** may continuously monitor the position of the dispensing end **166** within the received image data so that the position may be adjusted in real-time to the patient's eye **180** during the ophthalmic procedure; however in other embodiments, the surgical console controller **406** periodically detects the position of the dispensing end **166** within the received image data according to a predetermined schedule or time interval so that the position is only periodically checked/adjusted during selected moments of the ophthalmic procedure.

[0063] In certain embodiments, instead of automatically adjusting a position of the dispensing end **166** in response to detecting a change of relative position, the surgical console controller **406** can provide a notification to the surgeon or other operating staff to adjust the position of the lubrication assembly **104**. For example, in certain embodiments, the surgical console controller **406** may generate and transmit a visual signal to the display **112** to display a graphic or text message for inspection by the surgeon or other operating staff. In certain embodiments, the surgical console controller **406** may generate and transmit an audio signal to a speaker of the surgical console **120**, surgical microscope **102**, or lubrication assembly **104** to emit an audible tone or message for the surgeon or other operating staff.

[0064] In certain embodiments, the components and/or processing described with reference to FIG. **4** may also be included in, and/or performed by other devices in the surgical settings other than the surgical microscope **102**, the surgical microscope system **110**, and/or the surgical console **120**.

[0065] FIG. **5** is a flow diagram illustrating a technique for lubricating an eye of a patient (e.g., the eye **180** of the patient **130**) during an ophthalmic procedure, according to certain embodiments.

[0066] The technique begins at block 502. At block 502, a lubrication assembly (e.g., lubrication assembly 104) is moved into a selected position relative to the eye 180 of the patient 130. The lubrication assembly 104 may be moved into the selected position by moving the surgical microscope 102 and/or by actuating the movement actuator 154 operatively coupled to the lubrication assembly 104. The lubrication assembly 104 is moved to the selected position over the eye 180 such that the dispensing end 166 is disposed over the eye 180.

[0067] At block 504, a lubricant (e.g., lubricant 190) is dispensed from the lubrication assembly 104 onto the eye 180 of the patient 130, as described above with reference to various features in FIGS. 1-4. Generally, the lubricant 190 is dispensed from the dispensing end 166 of the dispenser 142 of the lubrication assembly 104. In certain embodiments, the lubricant 190 may be flowed from the lubricant source 146 to the dispenser 142 via the fluid line 144. For example, the lubricant 190 may be dispensed from the dispenser 142 by activating the lubricant actuator 148 as described with reference to FIG. 2, or by gravity feed. When the lubricant 190 is dispensed using the lubricant actuator 148, a dispense rate of the lubricant 190 is controlled via the lubricant actuator 148 and/or the controlled release valve 151. When the lubricant 190 is dispensed using gravity feed, the dispense rate of the lubricant 190 is controlled via the controlled release valve 151.

[0068] At block 506, a position of the lubrication assembly 104 (e.g., dispensing end 166) relative to the eye 180 of the patient 130 and/or a lubrication level of the eye 180 is monitored for a duration of the ophthalmic procedure. The position of the dispensing end 166 relative to the eye 180 and/or the lubrication level of the eye 180 may be monitored using digital signals corresponding to images captured by the camera 160. The digital signals may be communicated to a controller (e.g., the lubrication assembly controller 402, the surgical microscope controller 404, and/or the surgical console controller 406) for further processing and/or analysis.

[0069] In certain embodiments, a position of the lubrication assembly 104 (i.e., the dispensing end 166) relative to the eye 180 is determined from the digital signals, in order to determine whether the position of the dispensing end 166 should be adjusted. That is, the position of the dispensing end 166 and/or the eye 180 is monitored to ensure consistent dispensing of the lubricant 190 at a desired area of the eye 180. In certain embodiments, a lubrication or moisture level of the eye is determined from the digital signals, in order to determine whether a dispensing rate or the position of the dispensing end 166 should be adjusted. That is, the lubrication level of the eye 180 is monitored to determine whether to adjust the dispense rate or to move the lubrication assembly 104 to dispense onto a different area of the eye 180 to ensure sufficient lubrication of the eye 180.

[0070] At block 508a, in response to the monitoring at block 506, a message (e.g., a notification, a prompt, etc.) is generated and provided to a user (e.g., a surgeon or other operating staff), which may recommend that the user adjust the dispense rate of the lubrication assembly 104 or the position of the lubrication assembly 104. In certain embodiments, the message is provided to the user if the position of the eye 180 changes, the eye 180 or an area thereof needs more lubricant, and/or the lubrication assembly 104 is moved. Examples of a message may include indications of changes in lubrication level (of the eye 180 or area thereof), indications of changes in position, or indications that affirmatively instruct or prompt the surgeon, or operating staff to adjust the dispense rate and/or the position of the dispensing end 166. In certain embodiments, whether the message is provided to the user is determined by at least one of the controllers 402, 404, 406, which process images received via the camera 160.

[0071] At block 508b, in addition or alternatively to block 508a, a message (e.g., signal, instruction, etc.) is generated and provided to a corresponding controller in communication with the lubrication assembly 104 (e.g., the lubrication assembly controller 402, the surgical microscope controller 404, and/or the surgical console controller 406) to cause the controller to automatically adjust the dispense rate of the lubrication assembly 104 or the position of the lubrication assembly 104. For example, in certain embodiments, the dispense rate or the position of the dispensing end 166 is automatically adjusted by the lubrication assembly controller 402 if the eye 180 or an area

thereof needs more lubricant, the position of the eye **180** changes, and/or the lubrication assembly **104** is moved, based on a message provided to the lubrication assembly controller **402** in response to the monitoring at block **506**. In such embodiments, the corresponding controller (e.g., the lubrication assembly controller **402**, the surgical microscope controller **404**, and/or the surgical console controller **406**) sends controlling signals to the lubricant actuator **148** and/or the movement actuator **154** to adjust the dispense rate and/or the position of the dispensing end **166**.

[0072] The present disclosure may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the present disclosure is, therefore, indicated by the appended Claims rather than by this Detailed Description. All changes which come within the meaning and range of equivalency of the Claims are to be embraced within their scope.

[0073] Reference throughout this specification to features, advantages, or similar language does not imply that all the features and advantages that may be realized with the present disclosure should be or are in any single embodiment of the disclosure. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage, or characteristic described in connection with an embodiment is included in at least one embodiment of the present disclosure. Thus, discussions of the features and advantages, and similar language, throughout this specification may, but do not necessarily, refer to the same embodiment.

[0074] Furthermore, the described features, advantages, and characteristics of the disclosure may be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize, in light of the description herein, that the disclosure can be practiced without one or more of the specific features or advantages of a particular embodiment. In other instances, additional features and advantages may be recognized in certain embodiments that may not be present in all embodiments of the present disclosure.

[0075] Reference throughout this specification to “one embodiment”, “an embodiment”, or similar language means that a particular feature, structure, or characteristic described in connection with the indicated embodiment is included in at least one embodiment of the present disclosure. Thus, the phrases “in one embodiment”, “in an embodiment”, and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0076] The foregoing description is provided to enable any person skilled in the art to practice the various embodiments described herein. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments. Thus, the claims are not intended to be limited to the embodiments shown herein but are to be accorded the full scope consistent with the language of the claims.

Claims

1. A lubrication assembly for lubricating an eye of a patient during an ophthalmic procedure, the lubrication assembly comprising: a mounting bracket configured to be coupled to a surgical microscope; a dispenser coupled to the mounting bracket; a lubricant source fluidly coupled to the dispenser; a fluid line for movement of a lubricant between the lubricant source and the dispenser; and a first actuator for dispensing, from the dispenser, the lubricant onto the eye of the patient.
2. The lubrication assembly of claim 1, wherein the lubricant is automatically dispensed from the dispenser at a target dispense rate.
3. The lubrication assembly of claim 2, wherein the target dispense rate is based on a type of the ophthalmic procedure performed.
4. The lubrication assembly of claim 2, wherein the target dispense rate is set by a user.
5. The lubrication assembly of claim 2, wherein the target dispense rate is controlled by a controlled release valve coupled to the dispenser.
6. The lubrication assembly of claim 1, wherein: the dispenser is disposed in a field of view (FOV)

- of the surgical microscope; and the dispenser is outside of a focal plane of the surgical microscope.
- 7.** The lubrication assembly of claim 1 further comprising: a second actuator for moving the dispenser into a selected position relative to the eye of the patient.
- 8.** The lubrication assembly of claim 1, wherein a dispensing end of the dispenser is positioned at a center of an FOV of the surgical microscope.
- 9.** The lubrication assembly of claim 1, wherein a dispensing end of the dispenser is positioned between the eye of the patient and the surgical microscope.
- 10.** The lubrication assembly of claim 1, wherein the lubricant is automatically dispensed from the dispenser upon actuation of the actuator.
- 11.** A method for lubricating an eye of a patient during an ophthalmic procedure comprising: moving a lubrication assembly into a selected position relative to the eye of the patient; dispensing a lubricant from the lubrication assembly onto the eye of the patient; monitoring a position of the lubrication assembly relative to the eye of the patient and a lubrication level of the eye of the patient for a duration of the ophthalmic procedure; and generating a message to adjust a dispense rate of the lubrication assembly or the position of the lubrication assembly in response to the monitoring.
- 12.** The method of claim 11, wherein dispensing the lubricant comprises: automatically dispensing the lubricant from the lubrication assembly at a target dispense rate.
- 13.** The method of claim 11, wherein moving the lubrication assembly comprises: actuating an actuator coupled to the lubrication assembly to move a dispensing end of the lubrication assembly into the selected position.
- 14.** The method of claim 11, wherein moving the lubrication assembly comprises: actuating a surgical microscope comprising the lubrication assembly into the selected position.
- 15.** The method of claim 11, wherein monitoring the position of the lubrication assembly comprises: capturing an image of the patient's eye; determining a position of the patient's eye from the image; comparing the position of the lubrication assembly to the position of the patient's eye; and determining whether to move the lubrication assembly based on the comparison.
- 16.** The method of claim 11, the method further comprising: providing the message to a user of the lubrication assembly.
- 17.** The method of claim 11, the method further comprising: providing the message to a controller of the lubrication assembly, wherein providing the message to the controller of the lubrication assembly causes the controller to automatically adjust the dispense rate of the lubrication assembly or the position of the lubrication assembly.
- 18.** A method for lubricating an eye of a patient during an ophthalmic procedure comprising: moving a lubrication assembly into a selected position relative to the eye of the patient; dispensing a lubricant from the lubrication assembly onto the eye of the patient; monitoring a position of the lubrication assembly relative to the eye of the patient and a lubrication level of the eye of the patient for a duration of the ophthalmic procedure; and automatically adjusting a dispense rate of the lubrication assembly or the position of the lubrication assembly in response to the monitoring.
- 19.** The method of claim 18, wherein monitoring the position of the lubrication assembly relative to the eye of the patient and the lubrication level of the eye of the patient comprises: determining the eye of the patient needs more of the lubricant.
- 20.** The method of claim 18, wherein monitoring the position of the lubrication assembly relative to the eye of the patient and the lubrication level of the eye of the patient comprises: determining an area of the eye of the patient needs more of the lubricant.
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